

Notes A Functorial Excursion Between Algebraic Geometry and Linear Logic

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Contents

1	Preliminary Definitions	1
2	Motivational Thoughts	2
3	On the category of rings	4
4	On the category of R -modules and modules	4
5	The category $\text{Mod}^\ominus$ of modules and retromorphisms	6
6	The category Mod_R is symmetric monoidal closed	7
7	The category Mod as a symmetric monoidal ringed category	8
8	The categories Alg and CoAlg of (co)commutative (co)algebras	9
9	Functor of Points, Ring-spaces and Presheaves of Modules	10
10	The category PshMod of Presheaves of modules and forward morphisms	11
11	The category $\text{PshMod}^\ominus$ of presheaf modules and backward morphisms	15
12	The category PshMod is symmetric monoidal above $[\text{Ring}, \text{Set}]$	18

1 Preliminary Definitions

Definition 1.1. Let $\mathcal{C}$ be a category, then $\mathcal{C}$ is a Grothendieck category if the following is satisfied:

- $\mathcal{C}$ is Abelian, that is,
 - $\mathcal{C}$ has a zero object,

- $\mathcal{C}$ has all binary biproducts,
- $\mathcal{C}$ has all kernels and cokernels,
- all monomorphisms and epimorphisms are normal.
- $\mathcal{C}$ admits a generator, that is, there is $G \in \mathcal{C}$ such that for every parallel pair of morphisms $f, g : X \rightarrow Y$, $f \circ e = g \circ e$ implies $f = g$ for any $e : G \rightarrow X$.
- Let I be a directed set and let $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$ be an exact sequence for each $i \in I$, then $0 \rightarrow \operatorname{colim} A_i \rightarrow \operatorname{colim} B_i \rightarrow \operatorname{colim} C_i \rightarrow 0$ is an exact sequence.

Definition 1.2.

- Let $P : \mathcal{E} \rightarrow \mathcal{B}$ be a functor, then a morphism $f : A \rightarrow B$ in $\mathcal{E}$ is *Cartesian* if for any morphism $g : C \rightarrow B$ in $\mathcal{E}$ and $w : P(C) \rightarrow P(A)$ such that

$$\begin{array}{ccc} P(C) & \xrightarrow{P(g)} & P(B) \\ & \searrow w & \nearrow P(f) \\ & P(A) & \end{array}$$

there exists a unique $\hat{w} : C \rightarrow A$ in $\mathcal{E}$ such that $g = f \circ \hat{w}$ and $P(\hat{w}) = w$:

$$\begin{array}{ccc} \mathcal{E} & & \\ P \downarrow & & \\ \mathcal{B} & & \end{array} \quad \begin{array}{ccc} C & \xrightarrow{\forall g} & B \\ & \searrow \exists! w & \nearrow f \\ & A & \end{array} \quad \begin{array}{ccc} P(C) & \xrightarrow{P(g)} & P(B) \\ & \searrow \forall w & \nearrow P(f) \\ & P(A) & \end{array}$$

- A functor $P : \mathcal{E} \rightarrow \mathcal{B}$ is a *Grothendieck fibration* if for all $A \in \mathcal{E}$ and morphisms $f_0 : B \rightarrow P(A)$ in $\mathcal{B}$ there is a Cartesian morphism $f : B \rightarrow A$ such that $P(f) = f_0$. $P : \mathcal{E} \rightarrow \mathcal{B}$ is an *opfibration*, if $P^{op} : \mathcal{E}^{op} \rightarrow \mathcal{B}^{op}$ is a fibration, and $P : \mathcal{E} \rightarrow \mathcal{B}$ is a *bifibration* if it is both a fibration and opfibration.

2 Motivational Thoughts

Let us start by recalling basic principles of algebraic geometry. The intuition is that a space X comes along with a topological space (X, Σ_X) such that every open set U is equipped with a set $\mathcal{O}_X(U)$ of local operations called *regular functions*. Such regular functions are typically defined as polynomials with division and each $\mathcal{O}_X(U)$ defines a commutative ring. In functorial terms, every space gives rise to a ringed space, that is, a topological space (X, Σ_X) equipped with a contravariant functor $\mathcal{O}_X : \Sigma_X^{op} \rightarrow \mathbf{Ring}$ from the set of opens Σ_X^{op}

with the reversed inclusion order. The functorial action of $\mathcal{O}_X$ describes the restriction functions

$$\mathcal{O}_X(V \subseteq U) : \mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$$

which takes an operation $f \in \mathcal{O}_X(U)$ defined on an open set U and restrict it to an operation

$$f|_V := \mathcal{O}_X(V \subseteq U)(f)$$

on an open subset $V \subseteq U$. The functorial understanding of the concept of a space is supported by the observation that every commutative ring R defines a ringed space:

$$\text{Spec } R = (\text{Spec } R, \mathcal{O}_{\text{Spec } R})$$

on the set of prime ideals $\mathfrak{p}$ of the commutative ring R . Such a ringed space is called the *affine scheme* associated to R . Every space X of the theory, called a *scheme*, is a patchwork of affine schemes $\text{Spec } R_i$ glued together using the language of sheaf theory.

Every scheme X comes equipped with a symmetric monoidal category $\mathbf{qcMod}_X$ of quasicoherent sheaf of $\mathcal{O}_X$ -modules. By a presheaf of $\mathcal{O}_X$ -modules M of a ringed space $(X, \mathcal{O}_X)$, we mean a contravariant functor

$$M : \Omega_X^{op} \rightarrow \mathbf{Ab}$$

where $\mathbf{Ab}$ is the category of Abelian groups such that each Abelian group $M(U)$ defines a $\mathcal{O}_X(U)$ -module. The Serre-Swann theorem states a vector bundle on a suitable ringed space $(X, \mathcal{O}_X)$ can be equivalently encoded as its sheaf of sections that defines a locally free and projective sheaf of $\mathcal{O}_X$ -modules. A sheaf of $\mathcal{O}_X$ is called quasicoherent when it is locally presentable, that is, it is locally the cokernel of a morphism of free modules. The category $\mathbf{qcMod}_X$ extends the category of vector bundles to define an Abelian category where kernels and direct images can be computed.

The idea is to associate a model of intuitionistic linear logic, where formulas are understood as generalised vector bundles and proofs are vector fields by building an appropriate linear-non-linear adjunction. To associate a model of linear logic with each scheme X , we take the category $\mathbf{PshMod}_X$ of presheafs of $\mathcal{O}_X$ -modules and their homomorphisms as the underlying symmetric monoidal closed category with tensor product $\otimes_X$, the structural sheaf $\mathcal{O}_X$ as unit, and $-\circ_X$ for internal homs. For the cartesian part of a linear-non-linear adjunction, we take the category $\mathbf{PshCoAlg}_X$ of cocommutative comonoids (or commutative coalgebras) in $\mathbf{PshMod}_X$. It is convenient to the approach based on the functor of points (see [EH00, I.4]). Spec defines a contravariant functor $\text{Spec} : \mathbf{Ring}^{op} \rightarrow \mathbf{Scheme}$ that induces the *nerve functor*

$$\mathbf{nerve} : \mathbf{Scheme} \rightarrow [\mathbf{Ring}, \mathbf{Set}]$$

associating to every commutative ring R the set of R -points in a scheme X defined as $\mathbf{Scheme}(\text{Spec } R, X)$. The nerve functor is fully faithful and it allows

one to see every scheme X as a covariant presheaf

$$\mathbf{nerve}(X) : \mathbf{Ring} \rightarrow \mathbf{Set}$$

So we move to schemes to **Ring**-spaces and associate a model of intuitionistic linear logic with each **Ring**-space X , where formulas are interpreted in $\mathbf{PshMod}_X$ as presheaves of $\mathcal{O}_X$ -module understood as generalised vector bundles on the **Ring**-space. The linear-non-linear adjunction (and therefore the exponential modality) is constructed by the Sweedler construction.

3 On the category of rings

Let $(\mathcal{A}, \otimes, \mathbb{1})$ be a symmetric monoidal category with reflexive coequalisers preserved by tensor products in every coordinate. For example, the category **Ab** of Abelian groups and homomorphisms. A ring is a monoid object (R, m, e) in $(\mathcal{A}, \otimes, \mathbb{1})$. In other words, a ring is a triple such that (R, m, e) consisting of an object R and morphisms $m : R \otimes R \rightarrow R$ and $e : \mathbb{1} \rightarrow R$ such that the following diagrams commute:

$$\begin{array}{ccc} R \otimes R \otimes R & \xrightarrow{m \otimes R} & R \otimes R \\ R \otimes m \downarrow & & \downarrow m \\ R \otimes R & \xrightarrow{m} & R \end{array} \quad \begin{array}{ccc} R \otimes R & \xrightarrow{m} & R \\ m \downarrow & \nearrow & \downarrow e \otimes R \\ R & \xrightarrow{R \otimes e} & R \otimes R \end{array}$$

A ring (R, m, e) is commutative is the following triangle commutes:

$$\begin{array}{ccc} R \otimes R & \xrightarrow{\gamma_{R,R}} & R \otimes R \\ & \searrow m & \swarrow m \\ & R & \end{array}$$

A *ring morphism* $u : (R, m_R, e_R) \rightarrow (S, m_S, e_S)$ is a morphism $u : R \rightarrow S$ in the underlying category $\mathcal{A}$ making the following diagrams commute:

$$\begin{array}{ccc} R \otimes R & \xrightarrow{u \otimes u} & S \otimes S \\ m_S \downarrow & & \downarrow m_S \\ R & \xrightarrow{u} & S \end{array} \quad \begin{array}{ccc} & \mathbb{1} & \\ e_R \swarrow & & \searrow e_S \\ R & \xrightarrow{u} & S \end{array}$$

The category $\mathbf{Ring}(\mathcal{A})$ of rings in $\mathcal{A}$ defined by the tensor product $R, S \mapsto R \otimes S$ in $\mathcal{A}$.

4 On the category of R -modules and modules

Given a ring R in a symmetric monoidal category $\mathcal{A}$, an *R -module* is a pair (M, act) where M is an object in $\mathcal{A}$ along with a morphism $\text{act} : R \otimes M \rightarrow M$

in $\mathcal{A}$ such that:

$$\begin{array}{ccc}
 R \otimes R \otimes M & \xrightarrow{m_R \otimes M} & R \otimes M \\
 R \otimes \text{act} \downarrow & & \downarrow \text{act} \\
 R \otimes M & \xrightarrow{\text{act}} & M
 \end{array}
 \qquad
 \begin{array}{ccc}
 & R \otimes M & \\
 e_R \otimes M \nearrow & & \searrow \text{act} \\
 M & \xlongequal{\quad} & M
 \end{array}$$

Equivalently, an R -module is an Eilenberg-Moore algebra for the monad $A \mapsto R \otimes A$. An R -module homomorphism $f : M \rightarrow N$ between two modules is a map f in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{array}{ccc}
 R \otimes M & \xrightarrow{R \otimes f} & R \otimes N \\
 \text{act}_M \downarrow & & \downarrow \text{act}_N \\
 M & \xrightarrow{f} & N
 \end{array}$$

$\mathbf{Mod}_R$ is the category of R -modules and their morphisms.

A *module* is a pair (R, M) where R is a ring and M is a module. A *module homomorphism* $(u, f) : (R, M) \rightarrow (S, N)$ is a ring morphism $u : R \rightarrow S$ and a morphism $f : M \rightarrow N$ such that

$$\begin{array}{ccc}
 R \otimes M & \xrightarrow{u \otimes f} & S \otimes N \\
 \text{act}_M \downarrow & & \downarrow \text{act}_N \\
 M & \xrightarrow{f} & N
 \end{array}$$

$\mathbf{Mod}$ is the category of modules and their morphisms.

There is a functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ transporting every module (R, M) to the underlying ring R and every morphism (u, f) to u . We will write

$$u : R \rightarrow S \models f : M \rightarrow N$$

for a module morphism $(u, f) : (R, M) \rightarrow (S, N)$. The intuition behind the notation is that every ring homomorphism $u : R \rightarrow S$ induces a “fibre” consisting of all module morphisms $(u, f) : (R, M) \rightarrow (S, N)$ living above and refining the ring morphism $u : R \rightarrow S$. The fibre of πR for a commutative ring is the category $\mathbf{Mod}_R$. The functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ defines a Grothendieck fibration: every pair consisting of a ring morphism $u : R \rightarrow S$ and of an S -module (N, act_N) induces an R -module noted

$$\mathbf{res}_u N = (N, \text{act}'_N)$$

with the action morphism act'_N defined as the composite:

$$R \otimes N \xrightarrow{u \otimes N} S \otimes N \xrightarrow{\text{act}_N} N$$

Moreover, the S -module with a module morphism:

$$u : R \rightarrow S \models \text{id}_N : \mathbf{res}_u N \rightarrow N$$

which is weakly Cartesian in the sense that every module morphism

$$u : R \rightarrow S \models f : M \rightarrow N$$

factors uniquely as

$$M \xrightarrow{(id_R, h)} \mathbf{res}_u N \xrightarrow{(u, id_N)} N$$

for a morphism $h : M \rightarrow \mathbf{res}_u N$ in the category $\mathbf{Mod}_R$. The unique solution h is equal to the original morphism $f : M \rightarrow N$ viewed as a morphism in $\mathbf{Mod}_R$. Given a ring morphism $u : R \rightarrow S$, the existence of a weakly Cartesian map for every S -module N ensures the existence of a functor

$$\mathbf{res}_u : \mathbf{Mod}_S \rightarrow \mathbf{Mod}_R$$

called restriction of scalar along u . Along with the fact that the composite of weakly Cartesian maps is weakly Cartesian we have that the forgetful functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ is a Grothendieck fibration.

Now we use that fact the category $\mathcal{A}$ has reflexive coequalisers, so one can show that $\mathbf{res}_u$ has a left adjoint

$$\mathbf{ext}_u : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$$

constructed as follows. Every ring homomorphism $u : R \rightarrow S$ induces an $R \otimes S$ -module notes as $R \otimes_u S$ defined as the reflexive coequaliser of the following diagram

$$\begin{array}{ccc} & \xrightarrow{m_{R \otimes S}} & \\ R \otimes R \otimes S & \xleftarrow{R \otimes e_R \otimes S} & R \otimes S \\ & \xrightarrow{(R \otimes m_S) \circ (R \otimes u \otimes S)} & \end{array}$$

computed in $\mathcal{A}$. Given three commutative rings R , S_1 and S_2 , we define the functor

$$\otimes_R : \mathbf{Mod}_{S_1 \otimes R} \times \mathbf{Mod}_{R \otimes S_2} \rightarrow \mathbf{Mod}_{S_1 \otimes S_2}$$

The two maps $\text{act}_M : M \otimes R \rightarrow M$ and $\text{act}_N : N \otimes R \rightarrow N$ are deduced by restriction of scalar from the $S_1 \otimes R$ -module structure of M and the $R \otimes S_2$ -module structure of N . The left adjunction functor $\mathbf{ext}_u$ is then defined the following way:

$$\mathbf{ext}_u : M \mapsto M \otimes_R (R \otimes_u S)$$

5 The category $\mathbf{Mod}^\ominus$ of modules and retromorphisms

Let $[M, N]$ denote the internal hom of M and N in $\mathcal{A}$. A *module retromorphism*

$$(u, f) : (S, N) \rightarrow (R, M)$$

is a pair (u, f) where $u : R \rightarrow S$ is a ring morphism and $f : N \rightarrow M$ is a morphism in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{array}{ccccc} R \otimes M & \xleftarrow{R \otimes f} & R \otimes N & \xrightarrow{u \otimes N} & S \otimes N \\ \text{act}_M \downarrow & & & & \downarrow \text{act}_N \\ M & \xleftarrow{f} & N & & \end{array}$$

The category $\mathbf{Mod}^\ominus$ has ring modules as objects and module retromorphisms. There is a forgetful functor $\pi : \mathbf{Mod}^\ominus \rightarrow \mathbf{Ring}$ transporting every retromorphism to the underlying ring morphism and π is also a Grothendieck fibration. π is also a bifibration. Let $u : R \rightarrow S$ be a ring morphism, then functor defined by coextension of scalar along u

$$\mathbf{coext}_u : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$$

transport every R -module (M, act_M) to the S -module $[S, M]_u$ defined as the coreflexive equaliser of the below diagram:

$$\begin{array}{ccc} & \xrightarrow{[u \otimes S, M] \circ [m_S, M]} & \\ [S, M] & \xleftarrow{[e_R \otimes M, M]} [R \otimes S, M] & \xrightarrow{[R \otimes S, \text{act}_M] \circ [R \otimes _, R \otimes _]} \\ & & \end{array}$$

Note that $[S, M]_u$ provides an internal description of the set of morphisms $f : S \rightarrow M$ making the following diagram commute:

$$\begin{array}{ccc} R \otimes S & \xrightarrow{R \otimes f} & R \otimes M \\ u \otimes S \downarrow & & \downarrow \text{act}_M \\ S \otimes S & & \\ m_S \downarrow & & \downarrow \\ S & \xrightarrow{f} & M \end{array}$$

or, equivalently, as the set of morphisms $f : \mathbf{res}_u S \rightarrow M$. Thus, every ring morphism $u : R \rightarrow S$ induces the following sequence of adjunctions:

$$\begin{array}{ccc} & \xrightarrow{\mathbf{coext}_u} & \\ \mathbf{Mod}_R & \xleftarrow{\mathbf{res}_u} \mathbf{Mod}_S & \xrightarrow{\mathbf{ext}_u} \\ & & \end{array}$$

6 The category $\mathbf{Mod}_R$ is symmetric monoidal closed

It is well known that the category $\mathbf{Mod}_R$ of modules over a ring R is symmetric monoidal closed. Its tensor product

$$\otimes_R : \mathbf{Mod}_R \times \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

transports every pair of R -modules (M, N) into the coreflexive equaliser of the following diagram calculated in $\mathcal{A}$:

$$\begin{array}{ccc} & \xrightarrow{\text{act}_M \otimes N} & \\ M \otimes R \otimes N & \xleftarrow{M \otimes e_R \otimes N} M \otimes N & \\ & \xrightarrow{M \otimes \text{act}_N} & \end{array}$$

In $\mathbf{Mod}_R$, the tensorial unit is the R itself seen as an R -module. The internal hom

$$[\cdot, \cdot]_R : \mathbf{Mod}_R^{op} \times \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

takes every pair of R -modules M and N to the coreflexive equaliser of the following diagram:

$$\begin{array}{ccc} & \xrightarrow{[\text{act}_M, N]} & \\ [M, N] & \xleftarrow{[e_R \otimes M, N]} [R \otimes M, N] & \\ & \xrightarrow{[R \otimes M, \text{act}_N] \circ [R \otimes _, R \otimes _]} & \end{array}$$

7 The category $\mathbf{Mod}$ as a symmetric monoidal ringed category

Let us consider the category $\mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod}$ defined by the pullback diagram (in $\mathbf{Cat}$?) as follows:

$$\begin{array}{ccc} \mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} & \longrightarrow & \mathbf{Mod} \\ \downarrow & & \downarrow \pi \\ \mathbf{Mod} & \xrightarrow{\pi} & \mathbf{Ring} \end{array}$$

The fibrewise tensor product

$$\otimes_{\mathbf{Mod}} : \mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} \rightarrow \mathbf{Mod}$$

transports every pair of modules (R, M) and (R, N) on the same monoid R to the R -module $(R, M \otimes_R N)$, so every pair of module homomorphisms:

$$\begin{array}{l} u : R \rightarrow S \models h_1 : M_1 \rightarrow N_1 \\ u : R \rightarrow S \models h_2 : M_2 \rightarrow N_2 \end{array}$$

above the same ring morphism $u : R \rightarrow S$ to the module homomorphism:

$$u : R \rightarrow S \models h_1 \otimes_u h_2 : M_1 \otimes_R M_2 \rightarrow N_1 \otimes_R N_2$$

where $h_1 \otimes_u h_2$ is a unique morphism making the following diagram commute:

$$\begin{array}{ccc} M_1 \otimes R \otimes M_2 & \xrightarrow{h_1 \otimes u \otimes h_2} & N_1 \otimes S \otimes N_2 \\ M_1 \otimes \text{act}_{M_2} \downarrow \downarrow \text{act}_{M_1} \otimes M_2 & & N_1 \otimes \text{act}_{N_2} \downarrow \downarrow \text{act}_{N_1} \otimes N_2 \\ M_1 \otimes M_2 & \xrightarrow{h_1 \otimes h_2} & N_1 \otimes N_2 \\ \text{quotient} \downarrow & & \downarrow \text{quotient} \\ M_1 \otimes_R M_2 & \xrightarrow{h_1 \otimes_u h_2} & N_1 \otimes_S N_2 \end{array}$$

The fibrewise unit $\mathbb{1}_{\mathbf{Mod}} : \mathbf{Ring} \rightarrow \mathbf{Mod}$ takes every commutative ring and converts it to itself viewed as a module. Categorically, one can understand that with the concept of a *ringed category*, which defined as a pair $(\mathcal{C}, \pi)$ where $\mathcal{C}$ is a category and $\pi : \mathcal{C} \rightarrow \mathbf{Ring}$ is a functor. The slice 2-category $\mathcal{C}/\mathbf{Ring}$ consists of ringed categories, fibrewise functions as 1-cells and natural transformations as 2-cells. The 2-category $\mathcal{C}/\mathbf{Ring}$ is Cartesian; the fibrewise product and the unit define a symmetric pseudomonoid structure on the ringed category $(\mathbf{Mod}, \pi)$ in $\mathcal{C}/\mathbf{Ring}$.

8 The categories $\mathbf{Alg}$ and $\mathbf{CoAlg}$ of (co)commutative (co)algebras

Let R be a commutative ring in $\mathcal{A}$, a *commutative R -algebra* is a commutative monoid in the symmetric monoidal category $\mathbf{Mod}_R$. A *commutative algebra* is a pair (R, A) where R is a commutative ring and A is a commutative R -algebra. An algebra morphism $(u, f) : (R, A) \rightarrow (S, B)$ is a pair (u, f) where u is a ring morphism and $f : A \rightarrow B$ is a module morphism such that the following diagrams commute:

$$\begin{array}{ccc} A \otimes_R A & \xrightarrow{f \otimes_u f} & B \otimes_S B \\ m_A \downarrow & & \downarrow m_B \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} 1_R & \xrightarrow{u} & 1_S \\ e_A \downarrow & & \downarrow e_B \\ A & \xrightarrow{f} & B \end{array}$$

Assume that for every commutative ring R the symmetric monoidal category $\mathbf{Mod}_R$ has free commutative monoids, which means that the forgetful functor

$$\mathbf{Forget}_R : \mathbf{Alg}_R \rightarrow \mathbf{Mod}_R$$

has a left adjoint

$$\mathbf{Sym}_R : \mathbf{Mod}_R \rightarrow \mathbf{Alg}_R$$

and the adjunction $\mathbf{Mod} \dashv \mathbf{Alg}$ is also fibrewise above $\mathbf{Ring}$.

Given a commutative ring R , a *cocommutative R -coalgebra* K is defined as a cocommutative comonoid in the symmetric monoidal category $\mathbf{Mod}_R$. Let us also assume that for any commutative ring R , $\mathbf{Mod}_R$ has cofree cocommutative comonoids, then the forgetful functor

$$\mathbf{Forget}_R : \mathbf{CoAlg}_R \rightarrow \mathbf{Mod}_R$$

has a right adjoint

$$\mathbf{Sym}_R : \mathbf{Mod}_R \rightarrow \mathbf{CoAlg}_R.$$

The above adjunction can be defined at a purely formal level without describing explicitly the cofree construction $\mathbf{CoFree}_R$ in each fibre $\mathbf{Mod}_R$. One can describe the construction of the above adjunction explicitly in many cases,

for example when $R = k$ is an algebraic closed field, see [Mur15]. One can also describe it for a general commutative ring R . First of all, the free commutative algebra $\mathbf{Sym}_R$ is calculated the following way with enough limits in the category $\mathbf{Mod}_R$:

$$\mathbf{Sym}_R : M \mapsto \bigoplus_{n < \omega} M \otimes_R^{sym} \dots \otimes_R^{sym} M$$

where $M \otimes_R^{sym} \dots \otimes_R^{sym} M$ is the symmetrised tensor product of M with itself n times. If $R = k$ is a field, then the Sweedler construction transports every k -vector space V into the commutative k -algebra defined the below way:

$$\mathbf{CoFree}_k(V) = (\mathbf{Sym}_k(V^*))^\circ$$

where V^* is the dual of V and $A \mapsto A^\circ$ is the Sweedler finite dual constuction mapping every R -algebra A into the R -coalgebra A° . In the case of a general commutative ring, one needs to use the construction from [Por15].

9 Functor of Points, Ring-spaces and Presheaves of Modules

We work with covariant presheaves X, Y on the category $\mathbf{Ring}$ of commutative rings that we call **Ring-spaces**. To every such space $X : \mathbf{Ring} \rightarrow \mathbf{Set}$ we associate its Grothendieck category $\mathbf{Points}(X)$ whose objects are pairs (R, x) with $x \in X(R)$ and whose morphisms $(R, x) \rightarrow (S, y)$ are ring morphisms $u : R \rightarrow S$ transporting the element $x \in X(R)$ to the element $y \in X(S)$ in the sense that $X(u)(x) = y$.

The category $\mathbf{Points}(X)$ comes with an obvious functor $\pi_X : \mathbf{Points}(X) \rightarrow \mathbf{Ring}$ such that $\pi_X : (R, x) \mapsto R$. π_X is called the functor of points of X and it defines a discrete Grothendieck opfibration above the category $\mathbf{Ring}$. A morphism $f : X \rightarrow Y$ between **Ring-spaces** may be equivalently defined as a functor:

$$f : \mathbf{Points}(X) \rightarrow \mathbf{Points}(Y)$$

making the following diagram commute:

$$\begin{array}{ccc} \mathbf{Points}(X) & \xrightarrow{f} & \mathbf{Points}(Y) \\ & \searrow \pi_X & \swarrow \pi_Y \\ & \mathbf{Ring} & \end{array}$$

The functor f is necessarily a discrete opfibration. The following **Ring-space**

$$\mathbf{Spec} \mathbb{Z} : R \mapsto \{*_R\}$$

is the terminal object in the presheaf category $[\mathbf{Ring}, \mathbf{Set}]$, and that its Grothendieck category is isomorphism to $\mathbf{Ring}$.

The following definition is adapted from [Ros11].

Definition 9.1. A *presheaf of modules* M on the **Ring**-space or an $\mathcal{O}_X$ -module M consists of the following data:

- For each point $(R, x) \in \mathbf{Points}(X)$, a module M_x over the commutative ring R ,
- Each map $u : (R, x) \rightarrow (S, y)$ in $\mathbf{Points}(X)$, a module morphism

$$u : R \rightarrow S \models \theta(u, x) : M_x \rightarrow N_y$$

lining over the ring morphism $u : R \rightarrow S$.

The map θ is required to satisfy the following functorial properties: the identity on the point (R, x) in $\mathbf{Points}(X)$ is transported to the identity morphism on the associated R -module

$$1_R \models \theta(1_{(R, x)}) = 1_{M_x}$$

and given two maps $(u, x) : (R, x) \rightarrow (S, y)$ and $(v, y) : (S, y) \rightarrow (T, z)$ in $\mathbf{Points}(X)$, one has:

$$v \circ u \models \theta((v, y) \circ (u, x)) = \theta(v, y) \circ \theta(u, x)$$

where the composite is calculated in the discrete opfibration $\mathbf{Points}(X) \rightarrow \mathbf{Ring}$.

There is the following characterisation:

Proposition 9.1. An $\mathcal{O}_X$ -module M is the same thing as a functor $M : \mathbf{Points}(X) \rightarrow \mathbf{Mod}$ making the following diagram commute:

$$\begin{array}{ccc} \mathbf{Points}(X) & \xrightarrow{M} & \mathbf{Mod} \\ & \searrow \pi_X \quad \swarrow \pi & \\ & \mathbf{Ring} & \end{array}$$

Note that every **Ring**-every space X comes with a presheaf of modules called *the structure presheaf* of X , and defined as the composite

$$\mathcal{O}_X : \mathbf{Points}(X) \xrightarrow{\pi_X} \mathbf{Ring} \xrightarrow{i_{\mathbf{Ring}}} \mathbf{Mod}$$

where the functor $i_{\mathbf{Ring}} : \mathbf{Ring} \rightarrow \mathbf{Mod}$ is the section of the forgetful functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ transporting every ring to itself viewed as a module.

10 The category **PshMod** of Presheaves of modules and forward morphisms

We construct the category **PshMod** of *presheaves of modules and forward morphisms* as follows. A *presheaf of modules* (X, M) is defined as a pair consisting of

a **Ring**-space $X : \mathbf{Ring} \rightarrow \mathbf{Set}$ along with a presheaf $\mathcal{O}_X$ -module M as above. A *forward morphism* between presheafs of modules $(f, \varphi) : (X, M) \rightarrow (Y, N)$ is defined as a morphism $f : X \rightarrow Y$ along with a natural transformation:

$$\begin{array}{ccc} \mathbf{Points}(X) & \xrightarrow{\mathbf{Points}(f)} & \mathbf{Points}(Y) \\ & \searrow M \quad \xRightarrow{\varphi} \quad \swarrow N & \\ & \mathbf{Mod} & \end{array}$$

φ is also required to vertical above **Ring**, i.e. the natural transformation obtained by composing

$$\pi \circ M \xRightarrow{\pi \circ \varphi} \pi \circ N \circ \mathbf{Points}(f)$$

is equal to the identity natural transformation. Equivalently, one can say that the following natural transformation

$$\pi_X \xRightarrow{1} \pi \circ M \xRightarrow{\pi \circ \varphi} \pi \circ N \circ \mathbf{Points}(f) \xRightarrow{1} \pi_Y \circ \mathbf{Points}(f)$$

is obtained by composing the three natural transformations depicted below:

$$\begin{array}{ccc} \mathbf{Points}(X) & \xrightarrow{\mathbf{Points}(f)} & \mathbf{Points}(Y) \\ & \searrow M \quad \xRightarrow{\varphi} \quad \swarrow N & \\ & \mathbf{Mod} & \\ \pi_X \swarrow & \downarrow \pi & \searrow \pi_Y \\ & \mathbf{Ring} & \end{array}$$

(Note: In the original diagram, the arrows from $\mathbf{Points}(X)$ and $\mathbf{Points}(Y)$ to $\mathbf{Mod}$ are labeled M and N respectively, and the arrows from $\mathbf{Mod}$ to $\mathbf{Ring}$ are labeled π_X and π_Y respectively. The arrow from $\mathbf{Mod}$ to $\mathbf{Ring}$ is labeled π . The arrows from $\mathbf{Points}(X)$ and $\mathbf{Points}(Y)$ to $\mathbf{Mod}$ are also labeled id .)

coincides with the identity natural transformation from π_X to $\pi_Y \circ f$. There is an obvious functor

$$\mathbf{p} : \mathbf{PshMod} \rightarrow [\mathbf{Ring}, \mathbf{Set}]$$

that transports every presheaf of modules (X, M) to its underlying **Ring**-space X and every forward morphism $(f, \varphi) : (X, M) \rightarrow (Y, N)$ to $f : X \rightarrow Y$ between **Ring**-spaces. We will further write

$$f : X \rightarrow Y \models \varphi : M \rightarrow N$$

for a forward morphism $(f, \varphi) : (X, M) \rightarrow (Y, N)$ between presheaf modules.

The functor $\mathbf{p}$ is a Grothendieck fibration every **Ring**-space morphism $f : X \rightarrow Y$ induces a functor

$$f^* : \mathbf{PshMod}_Y \rightarrow \mathbf{PshMod}_X$$

transporting every $\mathcal{O}_Y$ -module N into the $\mathcal{O}_X$ -module $N \circ \mathbf{Points}(f)$ obtained by precomposing with the discrete fibration $\mathbf{Points}(f)$ as depicted below:

$$\begin{array}{ccc}
 \mathbf{Points}(X) & \xrightarrow{\mathbf{Points}(f)} & \mathbf{Points}(Y) \\
 & \searrow \pi_X \quad \swarrow N \quad \swarrow \pi_Y & \\
 & \mathbf{Mod} & \\
 & \downarrow \pi & \\
 & \mathbf{Ring} &
 \end{array}$$

$\mathbf{p}$ is also a Grothendieck bifibration, but it is less straightforward. Let us also assume that $\mathbf{Ring}$ as well as $\mathbf{Mod}_R$ has small colimits. Such a property is the case for, say, $\mathcal{A} = \mathbf{Ab}$.

Proposition 10.1. For every morphism $f : X \rightarrow Y$ of $\mathbf{Ring}$ -spaces, there is a functor

$$f_! : \mathbf{PshMod}_X \rightarrow \mathbf{PshMod}_Y$$

which is left adjoint to f^* .

Proof. A proof of the statement based on a 2-categorical construction of the $\mathcal{O}_Y$ -module $f_!(M)$. One uses the fact that the functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ is a Grothendieck bifibration, in particular, a Grothendieck opfibration. This implies that $\mathbf{Mod}$ has small colimits, which are preserved by π . Suppose we are given the following diagram:

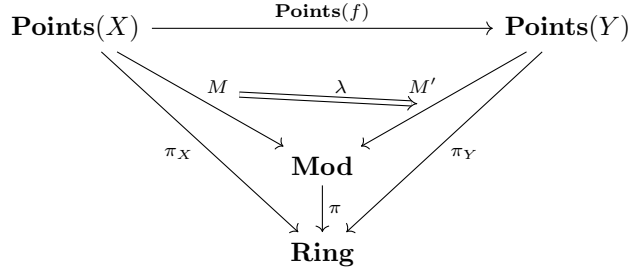
$$\begin{array}{ccc}
 \mathbf{Points}(X) & \xrightarrow{\mathbf{Points}(f)} & \mathbf{Points}(Y) \\
 & \searrow M \quad \swarrow \pi_Y & \\
 & \mathbf{Mod} & \\
 & \downarrow \pi & \\
 & \mathbf{Ring} &
 \end{array}$$

We start by calculating the left Kan extension of the functor M along the discrete fibration

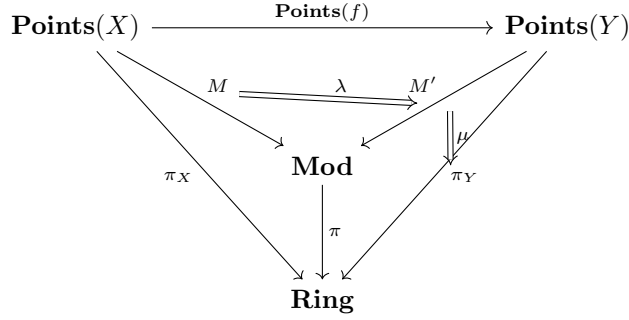
$$\mathbf{Points}(f) : \mathbf{Points}(X) \rightarrow \mathbf{Points}(Y)$$

One further obtains a functor $M' : \mathbf{Points}(Y) \rightarrow \mathbf{Mod}$ and a natural transformation λ exhibiting M' as the left Kan extension of M along points $\mathbf{Points}(f)$

depicted below:



The left Kan extension is a pointwise Kan extension calculated by small limits. As mentioned above, π preserves small colimits, and thus pointwise left Kan extensions. This established that the natural transformation $\pi \circ \lambda$ exhibits the functor $\pi \circ N'$ as the left Kan extension of the composite $\pi \circ M$ along $\mathbf{Points}(f)$. The universality property of left Kan extensions ensures that there exists a unique natural transformation $\mu : \pi \circ N' \Rightarrow \pi_Y$:



such that the following composite:

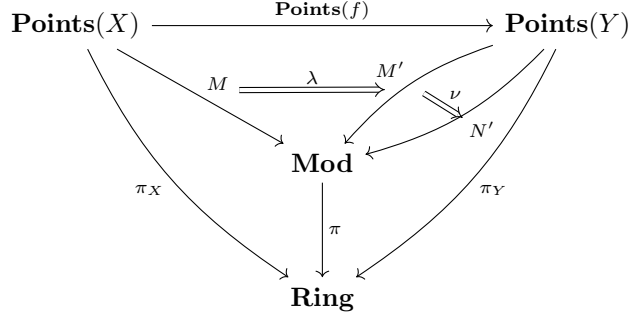
$$\pi \circ M \xrightarrow{\pi \circ \lambda} \pi \circ M' \circ \mathbf{Points}(f) \xrightarrow{\mu \circ \mathbf{Points}(f)} \pi_Y \circ \mathbf{Points}(f)$$

is equal to the identity natural transformation on the functor $\pi_X = \pi_Y \circ \mathbf{Points}(f)$. The functor π is a Grothendieck bifibration, so is the associated postcomposition functor:

$$\mathbf{Cat}(\mathbf{Points}(Y), \mathbf{Mod}) \rightarrow \mathbf{Cat}(\mathbf{Points}(Y), \mathbf{Ring})$$

is also a Grothendieck bifibration. In particular, there exists a functor $N : \mathbf{Points}(Y) \rightarrow \mathbf{Mod}$ and a natural transformation $\nu : M \Rightarrow N'$, which is co-cartesian above the natural transformation $\mu : \pi \circ M' \Rightarrow \pi \circ N$ as in the diagram

below:



The natural transformation $\varphi = \nu \circ \lambda : M \Rightarrow N \circ \mathbf{Points}(f)$ is vertical. Besides, $\pi \circ \varphi = 1$ and $N \circ \pi = \pi_Y$. The $\mathcal{O}_X$ -module N is defined $N : \mathbf{Points}(Y) \rightarrow \mathbf{Ring}$. Therefore we obtain a functor $f_! : \mathbf{PshMod}_X \rightarrow \mathbf{PshMod}_Y$ is left adjoint to the inverse image functor $f^* : \mathbf{PshMod}_Y \rightarrow \mathbf{PshMod}_X$ defined earlier. $\square$

Note that the $\mathcal{O}_Y$ -module $f_!(M)$ can be explicitly described by the following formula:

$$f_!(M) : y \in Y(R) \mapsto \bigoplus_{x \in X(R), fx=y} M_x \in \mathbf{Mod}_R$$

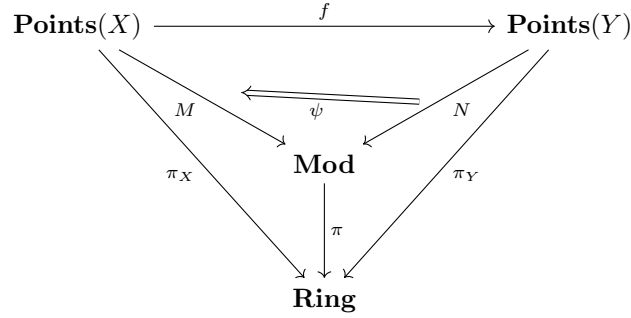
The adjunction $f_! \dashv f^*$ gives rise to a sequence of natural bijections formulated as the following derivations:

$$\frac{\frac{1_X : X \rightarrow X \models M \rightarrow f^*(N)}{f : X \rightarrow Y \models M \rightarrow N}}{1_Y : Y \rightarrow Y \models f_!(M) \rightarrow N}$$

11 The category $\mathbf{PshMod}^\ominus$ of presheaf modules and backward morphisms

We construct the category $\mathbf{PshMod}^\ominus$ of presheaves of modules and backwards morphisms. A *backward morphism* between presheaves of modules $(f, \psi) : (X, M) \rightarrow (Y, N)$ is a morphism $f : X \rightarrow Y$ between ringed spaces along with a natural transformation $\psi : N \circ f \Rightarrow M : \mathbf{Points}(X) \rightarrow \mathbf{Points}(Y)$ making the

following diagram commute:



The category $\mathbf{PshMod}^\ominus$ is the category of presheaves and their backward morphisms. There is an obvious functor

$$\mathbf{p}^{\ominus} : \mathbf{PshMod}^{\ominus} \rightarrow [\mathbf{Ring}, \mathbf{Set}]$$

so we write

$$f : X \rightarrow Y \models^{op} \psi : M \rightarrow N$$

for a backward morphism $(f, \psi) : (X, M) \rightarrow (Y, N)$ between presheaf modules. As the opposite of the fibration $\mathbf{p}$, the functor $\mathbf{p}^\ominus$ is also a Grothendieck fibration with the functor

$$(f^*)^{op} : \mathbf{PshMod}_Y^{op} \rightarrow \mathbf{PshMod}_X^{op}$$

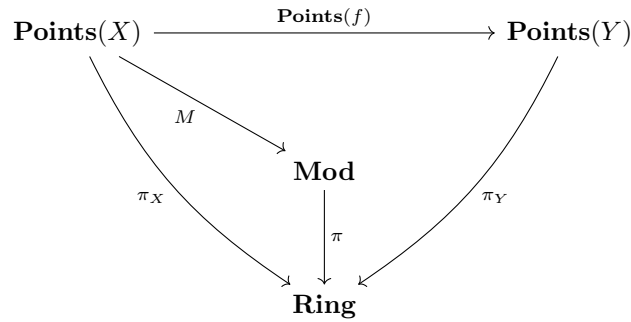
as a pullback functor. Let us make an extra assumption that both $\mathbf{Ring}$ and $\mathbf{Mod}_R$ have small limits.

Proposition 11.1. For every morphism $f : X \rightarrow Y$ between ringed spaces, there exists a functor called *direct image*

$$f_* : \mathbf{PshMod}_X \rightarrow \mathbf{PshMod}_Y$$

right adjoint to the functor f^* .

Proof. Suppose we are given the following diagram:



We then compute the right Kan extension of the discrete fibration functor

$$\mathbf{Points}(f) : \mathbf{Points}(X) \rightarrow \mathbf{Points}(Y)$$

Then we obtain a functor $M' : \mathbf{Points}(Y) \rightarrow \mathbf{Mod}$ and a natural transformation ρ exhibiting as the right extension of M along $\mathbf{Points}(f)$:

$$\begin{array}{ccc}
 \mathbf{Points}(X) & \xrightarrow{\mathbf{Points}(f)} & \mathbf{Points}(Y) \\
 & \swarrow M \quad \quad \quad \nwarrow M' & \\
 & \mathbf{Mod} & \\
 \pi_X \swarrow & \downarrow \pi & \searrow \pi_Y \\
 & \mathbf{Ring} &
 \end{array}$$

(Note: In the original image, a double arrow labeled ρ points from M' to M above the $\mathbf{Mod}$ node.)

Recall that the right Kan extension is a pointwise Kan extension computed by small limits. In turn, π preserves small limits, and, therefore, it preserves pointwise right Kan extensions. Thus, the natural transformation $\pi \circ \rho$ exhibits the functor $\pi \circ M'$ as the right Kan extension of the composite functor $\pi \circ M$ along $\mathbf{Points}(f)$. The universality property of right Kan extensions ensures that a natural transformation $\mu : \pi_Y \Rightarrow \pi \circ M'$ exists uniquely:

$$\begin{array}{ccc}
 \mathbf{Points}(X) & \xrightarrow{\mathbf{Points}(f)} & \mathbf{Points}(Y) \\
 & \swarrow M \quad \quad \quad \nwarrow M' & \\
 & \mathbf{Mod} & \\
 \pi_X \swarrow & \downarrow \pi & \searrow \pi_Y \\
 & \mathbf{Ring} &
 \end{array}$$

(Note: In the original image, a double arrow labeled μ points from π_Y to $\pi \circ M'$ above the $\mathbf{Ring}$ node.)

such that the composite

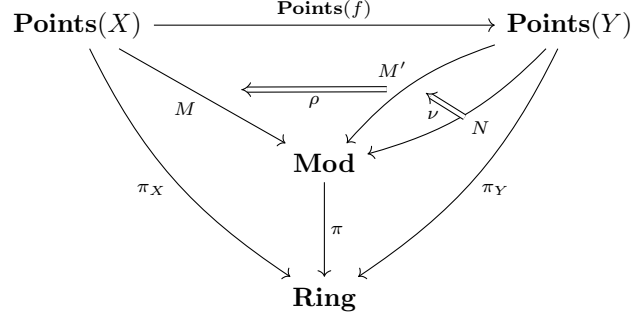
$$\pi_Y \circ \mathbf{Points}(f) \xrightarrow{\mu \circ \mathbf{Points}(f)} \pi \circ M' \circ \mathbf{Points}(f) \xrightarrow{\pi \circ \rho} \pi \circ M$$

is equal to the identity natural transformation on the functor $\pi_X = \pi_Y \circ \mathbf{Points}(f)$. Recall, that π is a Grothendieck bifibration, is so the postcomposition functor

$$\mathbf{Cat}(\mathbf{Points}(Y), \mathbf{Mod}) \rightarrow \mathbf{Cat}(\mathbf{Points}(Y), \mathbf{Ring}).$$

Thus, there is a functor $N : \mathbf{Points}(Y) \rightarrow \mathbf{Ring}$ and a natural transformation

$\nu : N \rightarrow M'$ such that



In particular, the natural transformation $\psi = \rho \circ \nu : N \circ \mathbf{Points}(f) \Rightarrow M$ is vertical. The $\mathcal{O}_Y$ -module $f_*(M)$ associated to the $\mathcal{O}_X$ -module N defined as $N : \mathbf{Points}(Y) \rightarrow \mathbf{Ring}$. Therefore, we obtain a functor $f_* : \mathbf{PshMod}_X \rightarrow \mathbf{PshMod}_Y$ which is right adjoint to $f^* : \mathbf{PshMod}_Y \rightarrow \mathbf{PshMod}_X$. $\square$

The functor $(f_*)^{op}$ is left adjoint to the functor $(f^*)^{op}$. The adjunction $f^* \dashv f_*$ gives rise to a sequence of natural bijections formulated in the following way:

$$\frac{1_X : X \rightarrow X \models^{op} M \rightarrow f^*(N)}{\frac{f : X \rightarrow Y \models^{op} M \rightarrow N}{1_Y : Y \rightarrow Y \models^{op} f_*(M) \rightarrow N}}$$

We conclude that every morphism $f : X \rightarrow Y$ between **Ring**-spaces induces three functors:

$$\begin{array}{ccc} & \xrightarrow{f_*} & \\ \mathbf{PshMod}_X & \xleftarrow{f^*} & \mathbf{PshMod}_X \\ & \xrightarrow{f_!} & \end{array}$$

organised into a sequence of adjunctions $f_! \dashv f^* \dashv f_*$.

12 The category **PshMod** is symmetric monoidal above **[Ring, Set]**

Recall that **[Ring, Set]** is Cartesian closed as a presheaf category. We exhibit the symmetric monoidal structure **PshMod** and show the functor $\mathbf{p} : \mathbf{PshMod} \rightarrow \mathbf{[Ring, Set]}$ is symmetric monoidal closed. The instruction is not quite common for algebraic geometry. The result establishes the general presheaves of modules as a foundations of connections between logic and algebraic geometry. Given a pair of **Ring**-space, $X, Y : \mathbf{Ring} \rightarrow \mathbf{Set}$ and a pair of presheaves of modules $M \in \mathbf{PshMod}_X$ and $N \in \mathbf{PshMod}_Y$. The cartesian

product $X \times Y$ of **Ring**-spaces is defined pointwise $X \times Y : R \mapsto R(X) \times R(Y)$. The tensor product $M \otimes N \in \mathbf{PshMod}_{X \times Y}$ is defined with the isomorphism

$$\mathbf{Points}(X \times Y) \cong \mathbf{Points}(X) \times_{\mathbf{Ring}} \mathbf{Points}(Y)$$

as the presheaves of modules

$$\mathbf{Points}(X \times Y) \xrightarrow{(M,N)} \mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} \xrightarrow{\otimes} \mathbf{Mod}$$

The unit of the tensor is defined as the structure presheaf of modules:

$$(\mathrm{Spec}(\mathbb{Z}), \mathcal{O}_{\mathrm{Spec}(\mathbb{Z})}) : (R, *_R) \mapsto R \in \mathbf{Mod}_R$$

on the terminal presheaf $\mathrm{Spec}(\mathbb{Z}) \in [\mathbf{Ring}, \mathbf{Set}]$, where $*_R$ is the singleton $\mathrm{Spec}(\mathbb{Z})(R)$. Recall that the internal hom $X \Rightarrow Y$ in $[\mathbf{Ring}, \mathbf{Set}]$ is the following covariant presheaf sending a commutative ring R to the set of natural transformations:

$$X \Rightarrow Y : R \mapsto ([\mathbf{Ring}, \mathbf{Set}]/\mathbf{y}_R)(\mathbf{y}_R \times X, \mathbf{y}_R \times Y)$$

making the following triangle commute:

$$\begin{array}{ccc} \mathbf{y}_R \times X & \xrightarrow{f} & \mathbf{y}_R \times Y \\ & \searrow \pi_{R,X} \quad \swarrow \pi_{R,Y} & \\ & \mathbf{y}_R & \end{array}$$

where $\mathbf{y}_R$ is the Yoneda presheaf generated by a commutative ring R in the following way:

$$\mathbf{y}_R : S \mapsto (R, S) : \mathbf{Ring} \rightarrow \mathbf{Set}$$

The presheaf of modules $M \multimap N \in \mathbf{PshMod}_{X \Rightarrow Y}$ is constructed in the following way. To every $f \in (X \Rightarrow Y)(R)$ we associate the R -module $(M \multimap N)_f$ consists of natural transformations φ making the following diagram commute:

$$\begin{array}{ccccc} \mathbf{Points}(\mathbf{y}_R \times X) & \xrightarrow{f} & \mathbf{Points}(\mathbf{y}_R \times Y) & & \\ \downarrow \mathbf{Points}(\pi_{R,X}) & \xRightarrow{\phi} & \downarrow \mathbf{Points}(\pi_{R,Y}) & & \\ \mathbf{Points}(X) & & \mathbf{Points}(Y) & & \\ & \searrow M \quad \swarrow N & & & \\ & \mathbf{Mod} & & & \\ & \downarrow \pi & & & \\ & \mathbf{Ring} & & & \end{array}$$

Such a natural transformation φ is a family of module morphisms

$$1_S : S \rightarrow S \models \varphi_{x,u} M_x \rightarrow N_{f(x,u)}$$

for $u : R \rightarrow S$ and $x \in X(S)$ natural in u and x in the sense that the following diagram commutes for any $v : S \rightarrow S'$ with $X(v)(x) = x'$ such that the following square commutes:

$$\begin{array}{ccc} M_x & \xrightarrow{\varphi_{x,u}} & N_{f(x,u)} \\ M_v \downarrow & & \downarrow N_{f(v,v)} \\ M_{x'} & \xrightarrow{\varphi_{x',v \circ u}} & N_{f(x',v \circ u)} \end{array}$$

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